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Circle

JEE Main Mathematics · Previous Year Questions with Answer Key

206	160	2002 – 2026	76	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical 31 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Consider the circle C : $x^2 + y^2 - 6x - 8y - 11 = 0$. Let a variable chord AB of the circle C subtend a right angle at the origin. If the locus of the foot of the perpendicular drawn from the origin on the chord AB is the circle $x^2 + y^2 - \alpha x - \beta y - \gamma = 0$, then $\alpha + \beta + 2\gamma$ is equal to ____ .

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let the line $x - y = 4$ intersect the circle $C : (x-4)^2 + (y+3)^2 = 9$ at the points Q and R . If $P(\alpha, \beta)$ is a point on C such that $PQ = PR$, then $(6\alpha + 8\beta)^2$ is equal to ____ .

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Let the centre of the circle $x^2 + y^2 + 2gx + 2fy + 25 = 0$ be in the first quadrant and lie on the line $2x - y = 4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x = 1$ is ____ .

Open solution

Q4. JEE Main 2026 (Online) 2nd April Morning Shift

Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line $x + y = 1$ is $\sqrt{14}$, then the square of the radius of C is _____.

Open solution

Q5. JEE Main 2026 (Online) 21st January Evening Shift

If P is a point on the circle $x^2 + y^2 = 4$, Q is a point on the straight line $5x + y + 2 = 0$ and $x - y + 1 = 0$ is the perpendicular bisector of PQ , then 13 times the sum of abscissa of all such points P is ____ .

Open solution

Q6. JEE Main 2025 (Online) 4th April Morning Shift

Let C be the circle $x^2 + (y-1)^2 = 2$, E_1 and E_2 be two ellipses whose centres lie at the origin and major axes lie on x -axis and y -axis respectively. Let the straight line $x + y = 3$ touch the curves C, E_1 and E_2 at $P(x_1, y_1), Q(x_2, y_2)$ and $R(x_3, y_3)$ respectively. Given that P is the mid point of the line segment QR and $PQ = \frac{2\sqrt{2}}{3}$, the value of $9(x_1y_1 + x_2y_2 + x_3y_3)$ is equal to _____.

Open solution

Q7. JEE Main 2025 (Online) 2nd April Morning Shift

The absolute difference between the squares of the radii of the two circles passing through the point $(-9, 4)$ and touching the lines $x + y = 3$ and $x - y = 3$, is equal to _____ .

Open solution

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Q8. JEE Main 2025 (Online) 23rd January Morning Shift

Let the circle C touch the line $x - y + 1 = 0$, have the centre on the positive x-axis, and cut off a chord of length $\frac{4}{\sqrt{13}}$ along the line $-3x + 2y = 1$. Let H be the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$, whose one of the foci is the centre of C and the length of the transverse axis is the diameter of C. Then $2\alpha^2 + 3\beta^2$ is equal to _____.

Open solution

Q9. JEE Main 2024 (Online) 9th April Morning Shift

Let the centre of a circle, passing through the points $(0, 0), (1, 0)$ and touching the circle $x^2 + y^2 = 9$, be (h, k). Then for all possible values of the coordinates of the centre (h, k), $4\left(h^2 + k^2\right)$ is equal to _____.

Open solution

Q10. JEE Main 2024 (Online) 30th January Evening Shift

Consider two circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - \alpha)^2 + y^2 = 16$, where $\alpha \in (5, 9)$. Let the angle between the two radii (one to each circle) drawn from one of the intersection points of C_1 and C_2 be $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$. If the length of common chord of C_1 and C_2 is β, then the value of $(\alpha\beta)^2$ equals _____.

Open solution

Q11. JEE Main 2024 (Online) 29th January Morning Shift

Equations of two diameters of a circle are $2x - 3y = 5$ and $3x - 4y = 7$. The line joining the points $\left(-\frac{22}{7}, -4\right)$ and $\left(-\frac{1}{7}, 3\right)$ intersects the circle at only one point $P(\alpha, \beta)$. Then, $17\beta - \alpha$ is equal to _____.

Open solution

Q12. JEE Main 2024 (Online) 27th January Evening Shift

Consider a circle $(x - \alpha)^2 + (y - \beta)^2 = 50$, where $\alpha, \beta > 0$. If the circle touches the line $y + x = 0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha + \beta)^2$ is equal to _____.

Open solution

Q13. JEE Main 2023 (Online) 12th April Morning Shift

Two circles in the first quadrant of radii r_1 and r_2 touch the coordinate axes. Each of them cuts off an intercept of 2 units with the line $x + y = 2$. Then $r_1^2 + r_2^2 - r_1 r_2$ is equal to _____.

Open solution

Q14. JEE Main 2023 (Online) 8th April Morning Shift

Consider a circle $C_1 : x^2 + y^2 - 4x - 2y = \alpha - 5$. Let its mirror image in the line $y = 2x + 1$ be another circle $C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$. Let r be the radius of C_2. Then $\alpha + r$ is equal to _____.

Open solution

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Q15. JEE Main 2023 (Online) 6th April Morning Shift

Let the point $(p, p+1)$ lie inside the region $E = \left\{(x, y) : 3 - x \leq y \leq \sqrt{9 - x^2}, 0 \leq x \leq 3\right\}$. If the set of all values of p is the interval (a, b), then $b^2 + b - a^2$ is equal to _____.

Open solution

Q16. JEE Main 2023 (Online) 6th April Morning Shift

A circle passing through the point $P(\alpha, \beta)$ in the first quadrant touches the two coordinate axes at the points A and B. The point P is above the line AB. The point Q on the line segment AB is the foot of perpendicular from P on AB. If PQ is equal to 11 units, then the value of $\alpha\beta$ is _____.

Open solution

Q17. JEE Main 2023 (Online) 30th January Evening Shift

Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C(\sqrt{2}, \sqrt{3})$. Let O be the origin and OC be perpendicular to both CP and CQ. If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$, then $a_1^2 + a_2^2 + b_1^2 + b_2^2$ is equal to :

Open solution

Q18. JEE Main 2022 (Online) 29th July Evening Shift

Let AB be a chord of length 12 of the circle $(x-2)^2 + (y+1)^2 = \frac{169}{4}$. If tangents drawn to the circle at points A and B intersect at the point P, then five times the distance of point P from chord AB is equal to _____.

Open solution

Q19. JEE Main 2022 (Online) 29th July Evening Shift

Let $S = \left\{(x, y) \in \mathbb{N} \times \mathbb{N} : 9(x-3)^2 + 16(y-4)^2 \leq 144\right\}$ and $T = \left\{(x, y) \in \mathbb{R} \times \mathbb{R} : (x-7)^2 + (y-4)^2 \leq 36\right\}$. Then $n(S \cap T)$ is equal to _____.

Open solution

Q20. JEE Main 2022 (Online) 29th July Morning Shift

Let the mirror image of a circle $c_1 : x^2 + y^2 - 2x - 6y + \alpha = 0$ in line $y = x + 1$ be $c_2 : 5x^2 + 5y^2 + 10gx + 10fy + 38 = 0$. If r is the radius of circle c_2, then $\alpha + 6r^2$ is equal to _____.

Open solution

Q21. JEE Main 2022 (Online) 28th June Evening Shift

If one of the diameters of the circle $x^2 + y^2 - 2\sqrt{2}x - 6\sqrt{2}y + 14 = 0$ is a chord of the circle $(x - 2\sqrt{2})^2 + (y - 2\sqrt{2})^2 = r^2$, then the value of r^2 is equal to _____.

Open solution

Q22. JEE Main 2022 (Online) 27th June Morning Shift

A rectangle R with end points of one of its sides as (1, 2) and (3, 6) is inscribed in a circle. If the equation of a diameter of the circle is 2x − y + 4 = 0, then the area of R is _____.

Open solution

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Q23. JEE Main 2022 (Online) 25th June Morning Shift

Let the abscissae of the two points P and Q be the roots of $2x^2 - rx + p = 0$ and the ordinates of P and Q be the roots of $x^2 - sx - q = 0$. If the equation of the circle described on PQ as diameter is $2(x^2 + y^2) - 11x - 14y - 22 = 0$, then $2r + s - 2q + p$ is equal to _____.

Open solution

Q24. JEE Main 2022 (Online) 24th June Evening Shift

Let a circle C : $(x - h)^2 + (y - k)^2 = r^2$, k > 0, touch the x-axis at (1, 0). If the line x + y = 0 intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of h + k + r is equal to _____.

Open solution

Q25. JEE Main 2021 (Online) 31st August Morning Shift

If the variable line 3x + 4y = α lies between the two
circles $(x - 1)^2 + (y - 1)^2 = 1$
and $(x - 9)^2 + (y - 1)^2 = 4$, without intercepting a chord on either circle, then the sum of all the integral values of α is _____.

Open solution

Q26. JEE Main 2021 (Online) 26th August Morning Shift

The locus of a point, which moves such that the sum of squares of its distances from the points (0, 0), (1, 0), (0, 1), (1, 1) is 18 units, is a circle of diameter d. Then d^2 is equal to _____.

Open solution

Q27. JEE Main 2021 (Online) 24th February Evening Shift

Let a point P be such that its distance from the point (5, 0) is thrice the distance of P from the point $(-5, 0)$. If the locus of the point P is a circle of radius r, then $4r^2$ is equal to _____

Open solution

Q28. JEE Main 2021 (Online) 24th February Morning Shift

If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C', whose center is at (2, 1), then its radius is _____.

Open solution

Q29. JEE Main 2020 (Online) 4th September Evening Slot

Let PQ be a diameter of the circle $x^2 + y^2 = 9$. If α and β are the lengths of the perpendiculars from P and Q on the straight line,
x + y = 2 respectively, then the maximum value of $\alpha\beta$ is _____.

Open solution

Q30. JEE Main 2020 (Online) 3rd September Morning Slot

The diameter of the circle, whose centre lies on the line x + y = 2 in the first quadrant and which touches both the lines x = 3 and y = 2, is _____ .

Open solution

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Q31. JEE Main 2020 (Online) 9th January Evening Slot

If the curves, $x^2 - 6x + y^2 + 8 = 0$ and
$x^2 - 8y + y^2 + 16 - k = 0$, $(k > 0)$ touch each other at a point, then the largest value of k is _____.

Open solution

MCQ (Single Correct Answer) 99 questions

Q32. JEE Main 2026 (Online) 6th April Evening Shift

Let C be a circle having centre in the first quadrant and touching the x-axis at a distance of 3 units from the origin. If the circle C has an intercept of length $6\sqrt{3}$ on y-axis, then the length of the chord of the circle C on the line $x - y = 3$ is :

(A) 8 (B) 6 (C) $6\sqrt{2}$ (D) $8\sqrt{2}$

Open solution

Q33. JEE Main 2026 (Online) 5th April Evening Shift

Let the point P be the vertex of the parabola $y = x^2 - 6x + 12$. If a line passing through the point P intersects the circle $x^2 + y^2 - 2x - 4y + 3 = 0$ at the points R and S , then the maximum value of $(\mathrm{PR} + \mathrm{PS})^2$ is :

(A) 10 (B) 20 (C) 25 (D) 5

Open solution

Q34. JEE Main 2026 (Online) 5th April Morning Shift

Let P be a moving point on the circle $x^2 + y^2 - 6x - 8y + 21 = 0$. Then, the maximum distance of P from the vertex of the parabola $x^2 + 6x + y + 13 = 0$ is equal to:

(A) 8 (B) 10 (C) 12 (D) 9

Open solution

Q35. JEE Main 2026 (Online) 4th April Morning Shift

Suppose that two chords, drawn from the point $(1, 2)$ on the circle $x^2 + y^2 + x - 3y = 0$ are bisected by the y-axis. If the other ends of these chords are R and S , and the mid point of the line segment RS is (α, β), then $6(\alpha + \beta)$ is equal to :

(A) 1 (B) 3 (C) 4 (D) 6

Open solution

Q36. JEE Main 2026 (Online) 2nd April Evening Shift

Let a circle pass through the origin and its centre be the point of intersection of two mutually perpendicular lines $x + (k - 1)y + 3 = 0$ and $2x + k^2y - 4 = 0$. If the line $x - y + 2 = 0$ intersects the circle at the points A and B, then $(AB)^2$ is equal to :

(A) 10 (B) 27 (C) 18 (D) 34

Open solution

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Q37. JEE Main 2026 (Online) 28th January Evening Shift

Let the circle $x^2 + y^2 = 4$ intersect x-axis at the points A$(a, 0)$, $a > 0$ and B$(b, 0)$. Let $P(2\cos\alpha, 2\sin\alpha)$, $0 < \alpha < \frac{\pi}{2}$ and $Q(2\cos\beta, 2\sin\beta)$ be two points such that $(\alpha - \beta) = \frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on :

Open solution

(A) $x^2 + y^2 - 4x - 4 = 0$

(B) $x^2 + y^2 - 4x - 4y = 0$

(C) $x^2 + y^2 - 4x - 4y - 4 = 0$

(D) $x^2 + y^2 - 4y - 4 = 0$

Q38. JEE Main 2026 (Online) 28th January Morning Shift

Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point $(2, 3)$ and is farthest from the center of C_2, is $x + ay + b = 0$, then $a - b$ is equal to

Open solution

(A) -6

(B) 10

(C) 6

(D) -2

Q39. JEE Main 2026 (Online) 24th January Morning Shift

Let a circle of radius 4 pass through the origin O , the points A$(-\sqrt{3}a, 0)$ and B$(0, -\sqrt{2}b)$, where a and b are real parameters and $ab \neq 0$. Then the locus of the centroid of $\triangle$OAB is a circle of radius

Open solution

(A) $\frac{7}{3}$

(B) $\frac{11}{3}$

(C) $\frac{5}{3}$

(D) $\frac{8}{3}$

Q40. JEE Main 2026 (Online) 22nd January Morning Shift

Let the set of all values of r, for which the circles $(x + 1)^2 + (y + 4)^2 = r^2$ and $x^2 + y^2 - 4x - 2y - 4 = 0$ intersect at two distinct points be the interval (α, β). Then $\alpha\beta$ is equal to

Open solution

(A) 21

(B) 24

(C) 20

(D) 25

Q41. JEE Main 2026 (Online) 21st January Morning Shift

Let PQ and MN be two straight lines touching the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ at the points A and B respectively. Let O be the centre of the circle and $\angle AOB = \pi/3$. Then the locus of the point of intersection of the lines PQ and MN is :

Open solution

(A) $x^2 + y^2 - 18x - 12y - 25 = 0$

(B) $x^2 + y^2 - 12x - 18y - 25 = 0$

(C) $3\left(x^2 + y^2\right) - 12x - 18y - 25 = 0$

(D) $3\left(x^2 + y^2\right) - 18x - 12y + 25 = 0$

Q42. JEE Main 2025 (Online) 7th April Morning Shift

Let C_1 be the circle in the third quadrant of radius 3 , that touches both coordinate axes. Let C_2 be the circle with centre $(1, 3)$ that touches C_1 externally at the point (α, β). If $(\beta - \alpha)^2 = \frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to

Open solution

(A) 22

(B) 13

(C) 9

(D) 31

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Q43. JEE Main 2025 (Online) 3rd April Evening Shift

If the four distinct points $(4, 6), (-1, 5), (0, 0)$ and $(k, 3k)$ lie on a circle of radius r, then $10k + r^2$ is equal to

(A) 34 (B) 32 (C) 35 (D) 33

Open solution

Q44. JEE Main 2025 (Online) 29th January Evening Shift

Let a circle C pass through the points (4, 2) and (0, 2), and its centre lie on 3x + 2y + 2 = 0. Then the length of the chord, of the circle C, whose mid-point is (1, 2), is:

(A) $4\sqrt{2}$ (B) $2\sqrt{2}$ (C) $2\sqrt{3}$ (D) $\sqrt{3}$

Open solution

Q45. JEE Main 2025 (Online) 29th January Morning Shift

Let the line $x + y = 1$ meet the circle $x^2 + y^2 = 4$ at the points A and B. If the line perpendicular to AB and passing through the mid-point of the chord AB intersects the circle at C and D, then the area of the quadrilateral ABCD is equal to :

(A) $\sqrt{14}$ (B) $3\sqrt{7}$ (C) $2\sqrt{14}$ (D) $5\sqrt{7}$

Open solution

Q46. JEE Main 2025 (Online) 28th January Morning Shift

Let the equation of the circle, which touches x-axis at the point $(a, 0), a > 0$ and cuts off an intercept of length b on y-axis be $x^2 + y^2 - \alpha x + \beta y + \gamma = 0$. If the circle lies below x-axis, then the ordered pair $(2a, b^2)$ is equal to

(A) $(\alpha, \beta^2 + 4\gamma)$ (B) $(\alpha, \beta^2 - 4\gamma)$
(C) $(\gamma, \beta^2 - 4\alpha)$ (D) $(\gamma, \beta^2 + 4\alpha)$

Open solution

Q47. JEE Main 2025 (Online) 24th January Morning Shift

Let circle C be the image of $x^2 + y^2 - 2x + 4y - 4 = 0$ in the line $2x - 3y + 5 = 0$ and A be the point on C such that OA is parallel to x-axis and A lies on the right hand side of the centre O of C. If $B(\alpha, \beta)$, with $\beta < 4$, lies on C such that the length of the arc AB is $(1/6)^{\text{th}}$ of the perimeter of C, then $\beta - \sqrt{3}\alpha$ is equal to

(A) $4 - \sqrt{3}$ (B) 3 (C) 4 (D) $3 + \sqrt{3}$

Open solution

Q48. JEE Main 2025 (Online) 22nd January Morning Shift

A circle C of radius 2 lies in the second quadrant and touches both the coordinate axes. Let r be the radius of a circle that has centre at the point $(2, 5)$ and intersects the circle C at exactly two points. If the set of all possible values of r is the interval (α, β), then $3\beta - 2\alpha$ is equal to :

(A) 10 (B) 12 (C) 14 (D) 15

Open solution

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Q49. JEE Main 2024 (Online) 9th April Morning Shift

Let a circle passing through $(2,0)$ have its centre at the point (h,k). Let $(x_\mathrm{c}, y_\mathrm{c})$ be the point of intersection of the lines $3x+5y=1$ and $(2+\mathrm{c})x+5\mathrm{c}^2y=1$. If $\mathrm{h}=\lim\limits_{\mathrm{c}\to 1} x_\mathrm{c}$ and $\mathrm{k}=\lim\limits_{\mathrm{c}\to 1} y_\mathrm{c}$, then the equation of the circle is :

Open solution

(A) $5x^2+5y^2-4x-2y-12=0$ (B) $25x^2+25y^2-20x+2y-60=0$

(C) $25x^2+25y^2-2x+2y-60=0$ (D) $5x^2+5y^2-4x+2y-12=0$

Q50. JEE Main 2024 (Online) 8th April Evening Shift

If the image of the point $(-4,5)$ in the line $x+2y=2$ lies on the circle $(x+4)^2+(y-3)^2=r^2$, then r is equal to:

Open solution

(A) 2 (B) 3 (C) 4 (D) 1

Q51. JEE Main 2024 (Online) 8th April Morning Shift

Let the circles $C_1:(x-\alpha)^2+(y-\beta)^2=r_1^2$ and $C_2:(x-8)^2+\left(y-\frac{15}{2}\right)^2=r_2^2$ touch each other externally at the point $(6,6)$. If the point $(6,6)$ divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio $2:1$, then $(\alpha+\beta)+4\left(r_1^2+r_2^2\right)$ equals

Open solution

(A) 130 (B) 110 (C) 145 (D) 125

Q52. JEE Main 2024 (Online) 6th April Evening Shift

If $\mathrm{P}(6,1)$ be the orthocentre of the triangle whose vertices are $\mathrm{A}(5,-2), \mathrm{B}(8,3)$ and $\mathrm{C}(\mathrm{h},\mathrm{k})$, then the point C lies on the circle :

Open solution

(A) $x^2+y^2-74=0$ (B) $x^2+y^2-65=0$ (C) $x^2+y^2-61=0$ (D) $x^2+y^2-52=0$

Q53. JEE Main 2024 (Online) 6th April Morning Shift

A circle is inscribed in an equilateral triangle of side of length 12. If the area and perimeter of any square inscribed in this circle are m and n, respectively, then $m+n^2$ is equal to

Open solution

(A) 408 (B) 414 (C) 312 (D) 396

Q54. JEE Main 2024 (Online) 5th April Evening Shift

Let the circle $C_1: x^2+y^2-2(x+y)+1=0$ and C_2 be a circle having centre at $(-1,0)$ and radius 2 . If the line of the common chord of C_1 and C_2 intersects the y-axis at the point P, then the square of the distance of P from the centre of C_1 is:

Open solution

(A) 4 (B) 6 (C) 2 (D) 1

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Q55. JEE Main 2024 (Online) 5th April Evening Shift

Let ABCD and AEFG be squares of side 4 and 2 units, respectively. The point E is on the line segment AB and the point F is on the diagonal AC. Then the radius r of the circle passing through the point F and touching the line segments BC and CD satisfies :

Open solution

(A) $\mathrm{r} = 1$ (B) $2\mathrm{r}^2 - 4\mathrm{r} + 1 = 0$ (C) $2\mathrm{r}^2 - 8\mathrm{r} + 7 = 0$ (D) $\mathrm{r}^2 - 8\mathrm{r} + 8 = 0$

Q56. JEE Main 2024 (Online) 5th April Morning Shift

Let a circle C of radius 1 and closer to the origin be such that the lines passing through the point $(3, 2)$ and parallel to the coordinate axes touch it. Then the shortest distance of the circle C from the point $(5, 5)$ is :

Open solution

(A) $4\sqrt{2}$ (B) 4 (C) 5 (D) $2\sqrt{2}$

Q57. JEE Main 2024 (Online) 4th April Evening Shift

Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line $x + y = 2$ intersect the circle C at the points P and Q. Let MN be a chord of C of length 2 unit and slope -1. Then, a distance (in units) between the chord PQ and the chord MN is

Open solution

(A) $3 - \sqrt{2}$ (B) $2 - \sqrt{3}$ (C) $\sqrt{2} - 1$ (D) $\sqrt{2} + 1$

Q58. JEE Main 2024 (Online) 4th April Morning Shift

A square is inscribed in the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. One side of this square is parallel to $y = x + 3$. If (x_i, y_i) are the vertices of the square, then $\Sigma\left(x_i^2 + y_i^2\right)$ is equal to:

Open solution

(A) 152 (B) 148 (C) 156 (D) 160

Q59. JEE Main 2024 (Online) 1st February Evening Shift

Let the locus of the midpoints of the chords of the circle $x^2 + (y - 1)^2 = 1$ drawn from the origin intersect the line $x + y = 1$ at P and Q. Then, the length of PQ is :

Open solution

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

Q60. JEE Main 2024 (Online) 1st February Morning Shift

Let $C : x^2 + y^2 = 4$ and $C' : x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C' intersect at two distinct points, is $\mathrm{R} - [\mathrm{a}, \mathrm{b}]$, then the point $(8\mathrm{a} + 12, 16\,\mathrm{b} - 20)$ lies on the curve :

Open solution

(A) $x^2 + 2y^2 - 5x + 6y = 3$ (B) $5x^2 - y = -11$

(C) $x^2 - 4y^2 = 7$ (D) $6x^2 + y^2 = 42$

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Q61. JEE Main 2024 (Online) 31st January Evening Shift

Let a variable line passing through the centre of the circle $x^2 + y^2 - 16x - 4y = 0$, meet the positive co-ordinate axes at the points A and B. Then the minimum value of $OA + OB$, where O is the origin, is equal to

(A) 12 (B) 20 (C) 24 (D) 18

Open solution

Q62. JEE Main 2024 (Online) 31st January Morning Shift

If one of the diameters of the circle $x^2 + y^2 - 10x + 4y + 13 = 0$ is a chord of another circle C, whose center is the point of intersection of the lines $2x + 3y = 12$ and $3x - 2y = 5$, then the radius of the circle C is :

(A) 4 (B) $3\sqrt{2}$ (C) 6 (D) $\sqrt{20}$

Open solution

Q63. JEE Main 2024 (Online) 30th January Morning Shift

If the circles $(x+1)^2 + (y+2)^2 = r^2$ and $x^2 + y^2 - 4x - 4y + 4 = 0$ intersect at exactly two distinct points, then

(A) $\frac{1}{2} < r < 7$ (B) $3 < r < 7$ (C) $5 < r < 9$ (D) $0 < r < 7$

Open solution

Q64. JEE Main 2024 (Online) 27th January Morning Shift

Four distinct points $(2k, 3k), (1, 0), (0, 1)$ and $(0, 0)$ lie on a circle for k equal to :

(A) $\frac{3}{13}$ (B) $\frac{2}{13}$ (C) $\frac{5}{13}$ (D) $\frac{1}{13}$

Open solution

Q65. JEE Main 2023 (Online) 10th April Evening Shift

Let A be the point $(1, 2)$ and B be any point on the curve $x^2 + y^2 = 16$. If the centre of the locus of the point P, which divides the line segment AB in the ratio $3 : 2$ is the point C(α, β), then the length of the line segment AC is :

(A) $\frac{3\sqrt{5}}{5}$ (B) $\frac{6\sqrt{5}}{5}$ (C) $\frac{2\sqrt{5}}{5}$ (D) $\frac{4\sqrt{5}}{5}$

Open solution

Q66. JEE Main 2023 (Online) 10th April Morning Shift

A line segment AB of length λ moves such that the points A and B remain on the periphery of a circle of radius λ. Then the locus of the point, that divides the line segment AB in the ratio 2 : 3, is a circle of radius :

(A) $\frac{2}{3}\lambda$ (B) $\frac{3}{5}\lambda$ (C) $\frac{\sqrt{19}}{7}\lambda$ (D) $\frac{\sqrt{19}}{5}\lambda$

Open solution

Q67. JEE Main 2023 (Online) 31st January Evening Shift

The set of all values of a^2 for which the line $x + y = 0$ bisects two distinct chords drawn from a point P $\left(\frac{1+a}{2}, \frac{1-a}{2}\right)$ on the circle $2x^2 + 2y^2 - (1+a)x - (1-a)y = 0$, is equal to :

(A) $(0, 4]$ (B) $(4, \infty)$ (C) $(2, 12]$ (D) $(8, \infty)$

Open solution

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Q68. JEE Main 2023 (Online) 29th January Morning Shift

Let the tangents at the points $A(4,-11)$ and $B(8,-5)$ on the circle $x^2+y^2-3x+10y-15=0$, intersect at the point C. Then the radius of the circle, whose centre is C and the line joining A and B is its tangent, is equal to :

Open solution

(A) $\frac{2\sqrt{13}}{3}$ (B) $\frac{3\sqrt{3}}{4}$ (C) $\sqrt{13}$ (D) $2\sqrt{13}$

Q69. JEE Main 2023 (Online) 25th January Morning Shift

The points of intersection of the line $ax+by=0,(a\neq b)$ and the circle $x^2+y^2-2x=0$ are $A(\alpha,0)$ and $B(1,\beta)$. The image of the circle with AB as a diameter in the line $x+y+2=0$ is :

Open solution

(A) $x^2+y^2+5x+5y+12=0$ (B) $x^2+y^2+3x+5y+8=0$

(C) $x^2+y^2-5x-5y+12=0$ (D) $x^2+y^2+3x+3y+4=0$

Q70. JEE Main 2023 (Online) 24th January Evening Shift

The locus of the mid points of the chords of the circle $C_1:(x-4)^2+(y-5)^2=4$ which subtend an angle θ_i at the centre of the circle C_1, is a circle of radius r_i. If $\theta_1=\frac{\pi}{3},\theta_3=\frac{2\pi}{3}$ and $r_1^2=r_2^2+r_3^2$, then θ_2 is equal to :

Open solution

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{6}$

Q71. JEE Main 2022 (Online) 28th July Morning Shift

For $\mathrm{t}\in(0,2\pi)$, if ABC is an equilateral triangle with vertices $\mathrm{A}(\sin t,-\cos \mathrm{t}),\mathrm{B}(\cos t,\sin t)$ and $C(a,b)$ such that its orthocentre lies on a circle with centre $\left(1,\frac{1}{3}\right)$, then $\left(a^2-b^2\right)$ is equal to :

Open solution

(A) $\frac{8}{3}$ (B) 8 (C) $\frac{77}{9}$ (D) $\frac{80}{9}$

Q72. JEE Main 2022 (Online) 28th July Morning Shift

Let C be the centre of the circle $x^2+y^2-x+2y=\frac{11}{4}$ and P be a point on the circle. A line passes through the point C, makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at the points Q and R. Then the area of the triangle PQR (in unit ^{2}) is :

Open solution

(A) 2 (B) $2\sqrt{2}$

(C) $8\sin\left(\frac{\pi}{8}\right)$ (D) $8\cos\left(\frac{\pi}{8}\right)$

Q73. JEE Main 2022 (Online) 27th July Morning Shift

If the circle $x^2+y^2-2gx+6y-19c=0,g,c\in\mathbb{R}$ passes through the point $(6,1)$ and its centre lies on the line $x-2cy=8$, then the length of intercept made by the circle on x-axis is :

Open solution

(A) $\sqrt{11}$ (B) 4 (C) 3 (D) $2\sqrt{23}$

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Q74. JEE Main 2022 (Online) 26th July Evening Shift

Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is _____.

Open solution

(A) 12 (B) 13 (C) 14 (D) 16

Q75. JEE Main 2022 (Online) 29th June Evening Shift

Let a triangle ABC be inscribed in the circle $x^2 - \sqrt{2}(x + y) + y^2 = 0$ such that $\angle BAC = \frac{\pi}{2}$. If the length of side AB is $\sqrt{2}$, then the area of the ΔABC is equal to :

Open solution

(A) 1 (B) $\left(\sqrt{6} + \sqrt{3}\right)/2$

(C) $\left(3 + \sqrt{3}\right)/4$ (D) $\left(\sqrt{6} + 2\sqrt{3}\right)/4$

Q76. JEE Main 2022 (Online) 27th June Evening Shift

The set of values of k, for which the circle $C : 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1, -\frac{1}{3}\right)$ lies on or inside the circle C, is :

Open solution

(A) an empty set (B) $\left(6, \frac{65}{9}\right]$

(C) $\left[\frac{80}{9}, 10\right)$ (D) $\left(9, \frac{92}{9}\right]$

Q77. JEE Main 2021 (Online) 26th August Evening Shift

A circle C touches the line x = 2y at the point (2, 1) and intersects the circle

$C_1 : x^2 + y^2 + 2y - 5 = 0$ at two points P and Q such that PQ is a diameter of C_1. Then the diameter of C is :

Open solution

(A) $7\sqrt{5}$ (B) 15 (C) $\sqrt{285}$ (D) $4\sqrt{15}$

Q78. JEE Main 2021 (Online) 26th August Morning Shift

If a line along a chord of the circle $4x^2 + 4y^2 + 120x + 675 = 0$, passes through the point $(-30, 0)$ and is tangent to the parabola $y^2 = 30x$, then the length of this chord is :

Open solution

(A) 5 (B) 7 (C) $5\sqrt{3}$ (D) $3\sqrt{5}$

Q79. JEE Main 2021 (Online) 27th July Evening Shift

Consider a circle C which touches the y-axis at (0, 6) and cuts off an intercept $6\sqrt{5}$ on the x-axis. Then the radius of the circle C is equal to :

Open solution

(A) $\sqrt{53}$ (B) 9 (C) 8 (D) $\sqrt{82}$

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Q80. JEE Main 2021 (Online) 27th July Morning Shift

Let P and Q be two distinct points on a circle which has center at C(2, 3) and which passes through origin O. If OC is perpendicular to both the line segments CP and CQ, then the set {P, Q} is equal to :

Open solution

(A) {(4, 0), (0, 6)}

(B) $\{(2+2\sqrt{2}, 3-\sqrt{5}), (2-2\sqrt{2}, 3+\sqrt{5})\}$

(C) $\{(2+2\sqrt{2}, 3+\sqrt{5}), (2-2\sqrt{2}, 3-\sqrt{5})\}$

(D) {(−1, 5), (5, 1)}

Q81. JEE Main 2021 (Online) 27th July Morning Shift

Let $A = \{(x, y) \in R \times R | 2x^2 + 2y^2 - 2x - 2y = 1\}$, $B = \{(x, y) \in R \times R | 4x^2 + 4y^2 - 16y + 7 = 0\}$ and $C = \{(x, y) \in R \times R | x^2 + y^2 - 4x - 2y + 5 \leq r^2\}$.

Open solution

Then the minimum value of |r| such that $A \cup B \subseteq C$ is equal to

(A) $\frac{3+\sqrt{10}}{2}$

(B) $\frac{2+\sqrt{10}}{2}$

(C) $\frac{3+2\sqrt{5}}{2}$

(D) $1+\sqrt{5}$

Q82. JEE Main 2021 (Online) 22th July Evening Shift

Let the circle S : $36x^2 + 36y^2 - 108x + 120y + C = 0$ be such that it neither intersects nor touches the co-ordinate axes. If the point of intersection of the lines, $x - 2y = 4$ and $2x - y = 5$ lies inside the circle S, then :

Open solution

(A) $\frac{25}{9} < C < \frac{13}{3}$

(B) 100 < C < 165

(C) 81 < C < 156

(D) 100 < C < 156

Q83. JEE Main 2021 (Online) 18th March Morning Shift

For the four circles M, N, O and P, following four equations are given :

Open solution

Circle M : $x^2 + y^2 = 1$

Circle N : $x^2 + y^2 - 2x = 0$

Circle O : $x^2 + y^2 - 2x - 2y + 1 = 0$

Circle P : $x^2 + y^2 - 2y = 0$

If the centre of circle M is joined with centre of the circle N, further center of circle N is joined with centre of the circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, then these lines form the sides of a :

(A) Rhombus

(B) Square

(C) Rectangle

(D) Parallelogram

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Q84. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, $(a < 0)$ be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to :

(A) $\sqrt{10}$ (B) $\sqrt{6}$ (C) $\sqrt{11}$ (D) $\sqrt{7}$

Q85. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Let A(1, 4) and B(1, -5) be two points. Let P be a point on the circle $(x - 1)^2 + (y - 1)^2 = 1$ such that $(PA)^2 + (PB)^2$ have maximum value, then the points, P, A and B lie on :

(A) a straight line (B) an ellipse (C) a parabola (D) a hyperbola

Q86. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

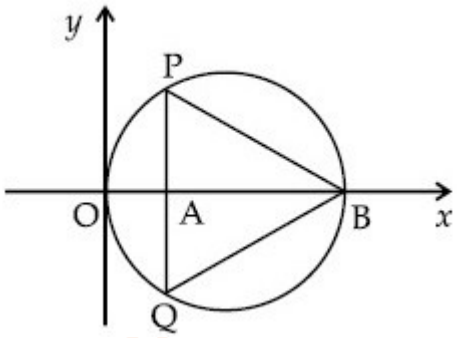
If the locus of the mid-point of the line segment from the point (3, 2) to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r, then r is equal to :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{3}$

Q87. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

In the circle given below, let OA = 1 unit, OB = 13 unit and PQ $\perp$ OB. Then, the area of the triangle PQB (in square units) is :



(A) $24\sqrt{2}$ (B) $24\sqrt{3}$ (C) $26\sqrt{2}$ (D) $26\sqrt{3}$

Q88. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

If the length of the chord of the circle, $x^2 + y^2 = r^2$ $(r > 0)$ along the line, $y - 2x = 3$ is r, then r^2 is equal to :

(A) $\frac{9}{5}$ (B) $\frac{24}{5}$ (C) $\frac{12}{5}$ (D) 12

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Q89. JEE Main 2019 (Online) 12th April Evening Slot

A circle touching the x-axis at (3, 0) and making an intercept of length 8 on the y-axis passes through the point :

(A) (1, 5) (B) (2, 3) (C) (3, 5) (D) (3, 10)

Open solution

Q90. JEE Main 2019 (Online) 12th April Morning Slot

If the angle of intersection at a point where the two circles with radii 5 cm and 12 cm intersect is 90°, then the length (in cm) of their common chord is :

(A) $\frac{13}{5}$ (B) $\frac{60}{13}$ (C) $\frac{120}{13}$ (D) $\frac{13}{2}$

Open solution

Q91. JEE Main 2019 (Online) 9th April Evening Slot

A rectangle is inscribed in a circle with a diameter lying along the line 3y = x + 7. If the two adjacent vertices of the rectangle are (−8, 5) and (6, 5), then the area of the rectangle (in sq. units) is :

(A) 72 (B) 84 (C) 56 (D) 98

Open solution

Q92. JEE Main 2019 (Online) 8th April Morning Slot

The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n$, $n \in N$, where N is the set of all natural numbers, is :

(A) 210 (B) 160 (C) 320 (D) 105

Open solution

Q93. JEE Main 2019 (Online) 12th January Evening Slot

If a circle of radius R passes through the origin O and intersects the coordinate axes at A and B, then the locus of the foot of perpendicular from O on AB is :

(A) $(x^2 + y^2)^2 = 4R^2x^2y^2$ (B) $(x^2 + y^2)(x + y) = R^2xy$

(C) $(x^2 + y^2)^2 = 4Rx^2y^2$ (D) $(x^2 + y^2)^3 = 4R^2x^2y^2$

Open solution

Q94. JEE Main 2019 (Online) 11th January Evening Slot

A circle cuts a chord of length 4a on the x-axis and passes through a point on the y-axis, distant 2b from the origin. Then the locus of the centre of this circle, is :

(A) an ellipse (B) a parabola (C) a hyperbola (D) a straight line

Open solution

Q95. JEE Main 2019 (Online) 11th January Morning Slot

The straight line x + 2y = 1 meets the coordinate axes at A and B. A circle is drawn through A, B and the origin. Then the sum of perpendicular distances from A and B on the tangent to the circle at the origin is :

(A) $4\sqrt{5}$ (B) $\frac{\sqrt{5}}{2}$ (C) $2\sqrt{5}$ (D) $\frac{\sqrt{5}}{4}$

Open solution

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Q96. JEE Main 2019 (Online) 11th January Morning Slot

A square is inscribed in the circle $x^2 + y^2 - 6x + 8y - 103 = 0$ with its sides parallel to the coordinate axes. Then the distance of the vertex of this square which is nearest to the origin is :

(A) $\sqrt{137}$ (B) 6 (C) $\sqrt{41}$ (D) 13

Open solution

Q97. JEE Main 2019 (Online) 10th January Evening Slot

If the area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq units then c is equal to :

(A) 20 (B) 25 (C) -25 (D) 13

Open solution

Q98. JEE Main 2018 (Online) 16th April Morning Slot

If a circle C, whose radius is 3, touches externally the circle, $x^2 + y^2 + 2x - 4y - 4 = 0$ at the point (2, 2), then the length of the intercept cut by this circle C, on the x-axis is equal to :

(A) $2\sqrt{5}$ (B) $3\sqrt{2}$ (C) $\sqrt{5}$ (D) $2\sqrt{3}$

Open solution

Q99. JEE Main 2017 (Online) 9th April Morning Slot

A line drawn through the point P(4, 7) cuts the circle $x^2 + y^2 = 9$ at the points A and B. Then PA·PB is equal to :

(A) 53 (B) 56 (C) 74 (D) 65

Open solution

Q100. JEE Main 2017 (Online) 9th April Morning Slot

The two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them is 60°. If the area of the quadrilateral is $4\sqrt{3}$, then the perimeter of the quadrilateral is :

(A) 12.5 (B) 13.2 (C) 12 (D) 13

Open solution

Q101. JEE Main 2017 (Online) 8th April Morning Slot

If a point P has co-ordinates (0, -2) and Q is any point on the circle, $x^2 + y^2 - 5x - y + 5 = 0$, then the maximum value of $(PQ)^2$ is :

(A) $\frac{25+\sqrt{6}}{2}$ (B) $14 + 5\sqrt{3}$ (C) $\frac{47+10\sqrt{6}}{2}$ (D) $8 + 5\sqrt{3}$

Open solution

Q102. JEE Main 2017 (Online) 8th April Morning Slot

If two parallel chords of a circle, having diameter 4units, lie on the opposite sides of the center and subtend angles $\cos^{-1}\left(\frac{1}{7}\right)$ and $\sec^{-1}(7)$ at the center respectivey, then the distance between these chords, is :

(A) $\frac{4}{\sqrt{7}}$ (B) $\frac{8}{\sqrt{7}}$ (C) $\frac{8}{7}$ (D) $\frac{16}{7}$

Open solution

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Q103. JEE Main 2016 (Online) 9th April Morning Slot

A circle passes through (−2, 4) and touches the y-axis at (0, 2). Which one of the following equations can represent a diameter of this circle?

(A) 4x + 5y − 6 = 0 (B) 2x − 3y + 10 = 0 (C) 3x + 4y − 3 = 0 (D) 5x + 2y + 4 = 0

Open solution

Q104. JEE Main 2016 (Offline)

If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at $(-3, 2)$, then the radius of S is :

(A) 5 (B) 10 (C) $5\sqrt{2}$ (D) $5\sqrt{3}$

Open solution

Q105. JEE Main 2015 (Online) 11th April Morning Slot

If the incentre of an equilateral triangle is (1, 1) and the equation of its one side is 3x + 4y + 3 = 0, then the equation of the circumcircle of this triangle is :

(A) $x^2 + y^2 - 2x - 2y - 2 = 0$ (B) $x^2 + y^2 - 2x - 2y - 14 = 0$

(C) $x^2 + y^2 - 2x - 2y + 2 = 0$ (D) $x^2 + y^2 - 2x - 2y - 7 = 0$

Open solution

Q106. JEE Main 2015 (Online) 11th April Morning Slot

If a circle passing through the point $(-1, 0)$ touches y-axis at (0, 2), then the length of the chord of the circle along the x-axis is :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) 3 (D) 5

Open solution

Q107. JEE Main 2015 (Online) 10th April Morning Slot

If y + 3x = 0 is the equation of a chord of the circle, $x^2 + y^2 - 30x = 0$, then the equation of the circle with this chord as diameter is :

(A) $x^2 + y^2 + 3x + 9y = 0$ (B) $x^2 + y^2 - 3x + 9y = 0$

(C) $x^2 + y^2 - 3x - 9y = 0$ (D) $x^2 + y^2 + 3x - 9y = 0$

Open solution

Q108. JEE Main 2015 (Offline)

Locus of the image of the point $(2, 3)$ in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0,\ k \in R$, is a :

(A) circle of radius $\sqrt{2}$. (B) circle of radius $\sqrt{3}$.

(C) straight line parallel to x-axis (D) straight line parallel to y-axis

Open solution

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Q109. JEE Main 2014 (Online) 9th April Morning Slot

If the point (1, 4) lies inside the circle $x^2 + y^2 - 6x - 10y + p = 0$ and the circle does not touch or intersect the coordinate axes, then the set of all possible values of p is the interval :

(A) (0, 25) (B) (25, 39) (C) (9, 25) (D) (25, 29)

Open solution

Q110. JEE Main 2013 (Online) 25th April Morning Slot

If a circle of unit radius is divided into two parts by an arc of another circle subtending an angle 60° on the circumference of the first circle, then the radius of the arc is:

(A) $\sqrt{3}$ (B) $\frac{1}{2}$ (C) 1 (D) $\sqrt{2}$

Open solution

Q111. JEE Main 2013 (Online) 23th April Morning Slot

If the circle $x^2 + y^2 - 6x - 8y + (25 - a^2) = 0$ touches the axis of x, then a equals.

(A) 0 (B) ± 4 (C) ± 2 (D) ± 3

Open solution

Q112. JEE Main 2013 (Online) 22th April Morning Slot

If two vertices of an equilateral triangle are A(-a, 0) and B(a, 0), a > 0, and the third vertex C lies above x-axis then the equation of the circumcircle of $\triangle ABC$ is:

(A) $3x^2 + 3y^2 - 2\sqrt{3}ay = 3a^2$ (B) $3x^2 + 3y^2 - 2ay = 3a^2$

(C) $x^2 + y^2 - 2ay = a^2$ (D) $x^2 + y^2 - \sqrt{3}ay = a^2$

Open solution

Q113. JEE Main 2013 (Online) 9th April Morning Slot

If each of the lines 5x + 8y = 13 and $4x - y = 3$ contains a diameter of the circle $x^2 + y^2 - 2(a^2 - 7a + 11)x - 2(a^2 - 6a + 6)y + b^3 + 1 = 0$, then

(A) a = 5 and $b \notin (-1, 1)$ (B) a = 1 and $b \notin (-1, 1)$

(C) a = 2 and $b \notin (-\infty, 1)$ (D) a = 5 and $b \in (-\infty, 1)$

Open solution

Q114. AIEEE 2012 (Online) 19th May

If the line $y = mx + 1$ meets the circle $x^2 + y^2 + 3x = 0$ in two points equidistant from and on opposite sides of x-axis, then

(A) $3m + 2 = 0$ (B) $3m - 2 = 0$ (C) $2m + 3 = 0$ (D) $2m - 3 = 0$

Open solution

Q115. AIEEE 2012 (Online) 19th May

If three distinct points A, B, C are given in the 2-dimensional coordinate plane such that the ratio of the distance of each one of them from the point $(1, 0)$ to the distance from $(-1, 0)$ is equal to $\frac{1}{2}$, then the circumcentre of the triangle ABC is at the point

(A) $\left(\frac{5}{3}, 0\right)$ (B) $(0, 0)$ (C) $\left(\frac{1}{3}, 0\right)$ (D) $(3, 0)$

Open solution

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Q116. AIEEE 2012

The length of the diameter of the circle which touches the x-axis at the point $(1, 0)$ and passes through the point $(2, 3)$ is :

(A) $\frac{10}{3}$ (B) $\frac{3}{5}$ (C) $\frac{6}{5}$ (D) $\frac{5}{3}$

Open solution

Q117. AIEEE 2010

The circle $x^2 + y^2 = 4x + 8y + 5$ intersects the line $3x - 4y = m$ at two distinct points if :

(A) $-35 < m < 15$ (B) $15 < m < 65$ (C) $35 < m < 85$ (D) $-85 < m < -35$

Open solution

Q118. AIEEE 2009

Three distinct points A, B and C are given in the 2 -dimensional coordinates plane such that the ratio of the distance of any one of them from the point $(1, 0)$ to the distance from the point $(-1, 0)$ is equal to $\frac{1}{3}$. Then the circumcentre of the triangle ABC is at the point :

(A) $\left(\frac{5}{4}, 0\right)$ (B) $\left(\frac{5}{2}, 0\right)$ (C) $\left(\frac{5}{3}, 0\right)$ (D) $(0, 0)$

Open solution

Q119. AIEEE 2008

The point diametrically opposite to the point $P(1, 0)$ on the circle $x^2 + y^2 + 2x + 4y - 3 = 0$ is :

(A) $(3, -4)$ (B) $(-3, 4)$ (C) $(-3, -4)$ (D) $(3, 4)$

Open solution

Q120. AIEEE 2006

If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π square units, the equation of the circle is :

(A) $x^2 + y^2 + 2x - 2y - 47 = 0$ (B) $x^2 + y^2 + 2x - 2y - 62 = 0$

(C) $x^2 + y^2 - 2x + 2y - 62 = 0$ (D) $x^2 + y^2 - 2x + 2y - 47 = 0$

Open solution

Q121. AIEEE 2006

Let C be the circle with centre $(0, 0)$ and radius 3 units. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its center is :

(A) $x^2 + y^2 = \frac{3}{2}$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = \frac{27}{4}$ (D) $x^2 + y^2 = \frac{9}{4}$

Open solution

Q122. AIEEE 2005

If the circles $x^2 + y^2 + 2ax + cy + a = 0$ and $x^2 + y^2 - 3ax + dy - 1 = 0$ intersect in two distinct points P and Q then the line 5x + by - a = 0 passes through P and Q for :

(A) exactly one value of a (B) no value of a

(C) infinitely many values of a (D) exactly two values of a

Open solution

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Q123. AIEEE 2005

If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = p^2$ orthogonally, then the equation of the locus of its centre is :

(A) $x^2 + y^2 - 3ax - 4by + (a^2 + b^2 - p^2) = 0$

(B) $2ax + 2by - (a^2 - b^2 + p^2) = 0$

(C) $x^2 + y^2 - 2ax - 3by + (a^2 - b^2 - p^2) = 0$

(D) $2ax + 2by - (a^2 + b^2 + p^2) = 0$

Open solution

Q124. AIEEE 2005

If the pair of lines $ax^2 + 2(a+b)xy + by^2 = 0$ lie along diameters of a circle and divide the circle into four sectors such that the area of one of the sectors is thrice the area of another sector then :

(A) $3a^2 - 10ab + 3b^2 = 0$

(B) $3a^2 - 2ab + 3b^2 = 0$

(C) $3a^2 + 10ab + 3b^2 = 0$

(D) $3a^2 + 2ab + 3b^2 = 0$

Open solution

Q125. AIEEE 2004

If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the locus of its centre is :

(A) $2ax - 2by - (a^2 + b^2 + 4) = 0$

(B) $2ax + 2by - (a^2 + b^2 + 4) = 0$

(C) $2ax - 2by + (a^2 + b^2 + 4) = 0$

(D) $2ax + 2by + (a^2 + b^2 + 4) = 0$

Open solution

Q126. AIEEE 2004

If the lines 2x + 3y + 1 = 0 and 3x - y - 4 = 0 lie along diameters of a circle of circumference 10π, then the equation of the circle is :

(A) $x^2 + y^2 + 2x - 2y - 23 = 0$

(B) $x^2 + y^2 - 2x - 2y - 23 = 0$

(C) $x^2 + y^2 + 2x + 2y - 23 = 0$

(D) $x^2 + y^2 - 2x + 2y - 23 = 0$

Open solution

Q127. AIEEE 2004

Intercept on the line y = x by the circle $x^2 + y^2 - 2x = 0$ is AB. Equation of the circle on AB as a diameter is :

(A) $x^2 + y^2 + x - y = 0$

(B) $x^2 + y^2 - x + y = 0$

(C) $x^2 + y^2 + x + y = 0$

(D) $x^2 + y^2 - x - y = 0$

Open solution

Q128. AIEEE 2003

The lines 2x - 3y = 5 and 3x - 4y = 7 are diameters of a circle having area as 154 sq. units. Then the equation of the circle is :

(A) $x^2 + y^2 - 2x + 2y = 62$

(B) $x^2 + y^2 + 2x - 2y = 62$

(C) $x^2 + y^2 + 2x - 2y = 47$

(D) $x^2 + y^2 - 2x + 2y = 47$

Open solution

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Q129. AIEEE 2002

If the chord y = mx + 1 of the circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle then value of m is :

(A) $2 \pm \sqrt{2}$ (B) $-2 \pm \sqrt{2}$ (C) $-1 \pm \sqrt{2}$ (D) none of these

Open solution

Q130. AIEEE 2002

The equation of a circle with origin as a center and passing through an equilateral triangle whose median is of length $3a$ is :

(A) $x^2 + y^2 = 9a^2$ (B) $x^2 + y^2 = 16a^2$ (C) $x^2 + y^2 = 4a^2$ (D) $x^2 + y^2 = a^2$

Open solution

Out of syllabus 76 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q131. JEE Main 2023 (Online) 29th January Evening Shift · Numerical

A circle with centre (2, 3) and radius 4 intersects the line $x + y = 3$ at the points P and Q. If the tangents at P and Q intersect at the point $S(\alpha, \beta)$, then $4\alpha - 7\beta$ is equal to _____.

Open solution

Q132. JEE Main 2023 (Online) 25th January Evening Shift · Numerical

Points P(−3, 2), Q(9, 10) and R(α, 4) lie on a circle C and PR as its diameter. The tangents to C at the points Q and R intersect at the point S. If S lies on the line $2x - ky = 1$, then k is equal to _____.

Open solution

Q133. JEE Main 2022 (Online) 25th July Evening Shift · Numerical

If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2\left(3 - \sqrt{3}\right)x + 2\left(4 - \sqrt{6}\right)y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$, touch internally at the point $P(\alpha, \beta)$, then $\left(\alpha + \sqrt{3}\right)^2 + \left(\beta + \sqrt{6}\right)^2$ is equal to _____.

Open solution

Q134. JEE Main 2022 (Online) 28th June Morning Shift · Numerical

Let the lines $y + 2x = \sqrt{11} + 7\sqrt{7}$ and $2y + x = 2\sqrt{11} + 6\sqrt{7}$ be normal to a circle $C : (x - h)^2 + (y - k)^2 = r^2$. If the line $\sqrt{11}y - 3x = \frac{5\sqrt{77}}{3} + 11$ is tangent to the circle C, then the value of $(5h - 8k)^2 + 5r^2$ is equal to _____.

Open solution

Q135. JEE Main 2022 (Online) 27th June Evening Shift · Numerical

Let a circle C of radius 5 lie below the x-axis. The line L_1 : 4x + 3y + 2 = 0 passes through the centre P of the circle C and intersects the line L_2 = 3x − 4y − 11 = 0 at Q. The line L_2 touches C at the point Q. Then the distance of P from the line 5x − 12y + 51 = 0 is _____.

Open solution

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Q136. JEE Main 2021 (Online) 31st August Evening Shift · **Numerical**

Let B be the centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$. Let the tangents at two points P and Q on the circle intersect at the point A(3, 1). Then $8.\left(\frac{area\ \Delta APQ}{area\ \Delta BPQ}\right)$ is equal to _____.

Open solution

Q137. JEE Main 2021 (Online) 27th August Evening Shift · **Numerical**

Two circles each of radius 5 units touch each other at the point (1, 2). If the equation of their common tangent is 4x + 3y = 10, and $C_1(\alpha, \beta)$ and $C_2(\gamma, \delta)$, $C_1 \neq C_2$ are their centres, then $|(\alpha + \beta)(\gamma + \delta)|$ is equal to _____.

Open solution

Q138. JEE Main 2021 (Online) 27th August Morning Shift · **Numerical**

Let the equation $x^2 + y^2 + px + (1 - p)y + 5 = 0$ represent circles of varying radius $r \in (0, 5]$. Then the number of elements in the set $S = \{q : q = p^2$ and q is an integer$\}$ is _____.

Open solution

Q139. JEE Main 2021 (Online) 17th March Morning Shift · **Numerical**

The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$x^2 + y^2 - 10x - 10y + 41 = 0$

$x^2 + y^2 - 24x - 10y + 160 = 0$ is _____.

Open solution

Q140. JEE Main 2021 (Online) 16th March Morning Shift · **Numerical**

Let ABCD be a square of side of unit length. Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the lines AD and AB are tangent to it, is also drawn. Let a tangent line from the point C to the circle C_2 meet the side AB at E. If the length of EB is $\alpha + \sqrt{3}\,\beta$, where α, β are integers, then $\alpha + \beta$ is equal to _____.

Open solution

Q141. JEE Main 2021 (Online) 24th February Evening Shift · **Numerical**

If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x - 2)^2 + (y - 3)^2 = 25$ at the point (5, 7) is A, then 24A is equal to _____.

Open solution

Q142. JEE Main 2020 (Online) 2nd September Morning Slot · **Numerical**

The number of integral values of k for which the line, 3x + 4y = k intersects the circle, $x^2 + y^2 - 2x - 4y + 4 = 0$ at two distinct points is _____.

Open solution

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Q143. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

The number of common tangents, to the circles

$x^2 + y^2 - 18x - 15y + 131 = 0$

and $x^2 + y^2 - 6x - 6y - 7 = 0$, is :

(A) 4 (B) 2 (C) 3 (D) 1

Open solution

Q144. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

Let the centre of a circle C be (α, β) and its radius $r < 8$. Let $3x + 4y = 24$ and $3x - 4y = 32$ be two tangents and $4x + 3y = 1$ be a normal to C. Then $(\alpha - \beta + r)$ is equal to :

(A) 7 (B) 9 (C) 5 (D) 6

Open solution

Q145. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let O be the origin and OP and OQ be the tangents to the circle $x^2 + y^2 - 6x + 4y + 8 = 0$ at the points P and Q on it. If the circumcircle of the triangle OPQ passes through the point $\left(\alpha, \frac{1}{2}\right)$, then a value of α is :

(A) 1 (B) $-\frac{1}{2}$ (C) $\frac{5}{2}$ (D) $\frac{3}{2}$

Open solution

Q146. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point $R\left(\frac{9}{4}, 2\right)$, then the area of the triangle PQR is :

(A) $\frac{13}{8}$ (B) $\frac{5}{8}$ (C) $\frac{5}{4}$ (D) $\frac{13}{4}$

Open solution

Q147. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

Let a circle C_1 be obtained on rolling the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ upwards 4 units on the tangent T to it at the point $(3, 2)$. Let C_2 be the image of C_1 in T. Let A and B be the centers of circles C_1 and C_2 respectively, and M and N be respectively the feet of perpendiculars drawn from A and B on the x-axis. Then the area of the trapezium AMNB is :

(A) $2\left(2 + \sqrt{2}\right)$ (B) $4\left(1 + \sqrt{2}\right)$ (C) $3 + 2\sqrt{2}$ (D) $2\left(1 + \sqrt{2}\right)$

Open solution

Q148. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

Let $y = x + 2, 4y = 3x + 6$ and $3y = 4x + 1$ be three tangent lines to the circle $(x - h)^2 + (y - k)^2 = r^2$. Then $h + k$ is equal to :

(A) 6 (B) 5 (1 + $\sqrt{2}$) (C) 5 (D) $5\sqrt{2}$

Open solution

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Q149. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

Let the tangents at two points A and B on the circle $x^2 + y^2 - 4x + 3 = 0$ meet at origin $O(0, 0)$. Then the area of the triangle OAB is :

(A) $\frac{3\sqrt{3}}{2}$ (B) $\frac{3\sqrt{3}}{4}$ (C) $\frac{3}{2\sqrt{3}}$ (D) $\frac{3}{4\sqrt{3}}$

Open solution

Q150. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y = 2x$ gives a chord OA of circle C_1. Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q, then QA : AP is equal to :

(A) 1 : 4 (B) 1 : 5 (C) 2 : 5 (D) 1 : 3

Open solution

Q151. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

Consider three circles:

$C_1 : x^2 + y^2 = r^2$

$C_2 : (x - 1)^2 + (y - 1)^2 = r^2$

$C_3 : (x - 2)^2 + (y - 1)^2 = r^2$

If a line L : y = mx + c be a common tangent to C_1, C_2 and C_3 such that C_1 and C_3 lie on one side of line L while C_2 lies on other side, then the value of $20(r^2 + c)$ is equal to :

(A) 23 (B) 15 (C) 12 (D) 6

Open solution

Q152. JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

Let the tangent to the circle $C_1 : x^2 + y^2 = 2$ at the point M$(-1, 1)$ intersect the circle $C_2 : (x - 3)^2 + (y - 2)^2 = 5$, at two distinct points A and B. If the tangents to C_2 at the points A and B intersect at N, then the area of the triangle ANB is equal to :

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{5}{3}$

Open solution

Q153. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

If the tangents drawn at the points $O(0, 0)$ and $P\left(1 + \sqrt{5}, 2\right)$ on the circle $x^2 + y^2 - 2x - 4y = 0$ intersect at the point Q, then the area of the triangle OPQ is equal to :

(A) $\frac{3+\sqrt{5}}{2}$ (B) $\frac{4+2\sqrt{5}}{2}$ (C) $\frac{5+3\sqrt{5}}{2}$ (D) $\frac{7+3\sqrt{5}}{2}$

Open solution

Q154. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let C be a circle passing through the points A$(2, -1)$ and B$(3, 4)$. The line segment AB is not a diameter of C. If r is the radius of C and its centre lies on the circle $(x - 5)^2 + (y - 1)^2 = \frac{13}{2}$, then r^2 is equal to :

(A) 32 (B) $\frac{65}{2}$ (C) $\frac{61}{2}$ (D) 30

Open solution

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Q155. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

A circle touches both the y-axis and the line x + y = 0. Then the locus of its center is :

(A) $y = \sqrt{2}x$ (B) $x = \sqrt{2}y$ (C) $y^2 - x^2 = 2xy$ (D) $x^2 - y^2 = 2xy$

Open solution

Q156. JEE Main 2022 (Online) 25th June Morning Shift · MCQ (Single Correct Answer)

Let a circle C touch the lines $L_1 : 4x - 3y + K_1 = 0$ and $L_2 : 4x - 3y + K_2 = 0$, $K_1, K_2 \in R$. If a line passing through the centre of the circle C intersects L_1 at $(-1, 2)$ and L_2 at $(3, -6)$, then the equation of the circle C is :

(A) $(x-1)^2 + (y-2)^2 = 4$ (B) $(x+1)^2 + (y-2)^2 = 4$

(C) $(x-1)^2 + (y+2)^2 = 16$ (D) $(x-1)^2 + (y-2)^2 = 16$

Open solution

Q157. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Two tangents are drawn from the point P(−1, 1) to the circle $x^2 + y^2 - 2x - 6y + 6 = 0$. If these tangents touch the circle at points A and B, and if D is a point on the circle such that length of the segments AB and AD are equal, then the area of the triangle ABD is equal to :

(A) 2 (B) $(3\sqrt{2} + 2)$ (C) 4 (D) $3(\sqrt{2} - 1)$

Open solution

Q158. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$, then a + b is equal to :

(A) 3 (B) 11 (C) 5 (D) 7

Open solution

Q159. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

Let $S_1 : x^2 + y^2 = 9$ and $S_2 : (x-2)^2 + y^2 = 1$. Then the locus of center of a variable circle S which touches S_1 internally and S_2 externally always passes through the points :

(A) $\left(\frac{1}{2}, \pm\frac{\sqrt{5}}{2}\right)$ (B) $(1, \pm 2)$

(C) $\left(2, \pm\frac{3}{2}\right)$ (D) $(0, \pm\sqrt{3})$

Open solution

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Q160. JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

Choose the correct statement about two circles whose equations are given below :

$x^2 + y^2 - 10x - 10y + 41 = 0$

$x^2 + y^2 - 22x - 10y + 137 = 0$

(A) circles have same centre

(B) circles have no meeting point

(C) circles have only one meeting point

(D) circles have two meeting points

Open solution

Q161. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

Let the tangent to the circle $x^2 + y^2 = 25$ at the point R(3, 4) meet x-axis and y-axis at points P and Q, respectively. If r is the radius of the circle passing through the origin O and having centre at the incentre of the triangle OPQ, then r^2 is equal to :

(A) $\frac{585}{66}$ (B) $\frac{625}{72}$ (C) $\frac{529}{64}$ (D) $\frac{125}{72}$

Open solution

Q162. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

Two tangents are drawn from a point P to the circle $x^2 + y^2 - 2x - 4y + 4 = 0$, such that the angle between these tangents is $\tan^{-1}\left(\frac{12}{5}\right)$, where $\tan^{-1}\left(\frac{12}{5}\right) \in (0, \pi)$. If the centre of the circle is denoted by C and these tangents touch the circle at points A and B, then the ratio of the areas of ΔPAB and ΔCAB is :

(A) 3 : 1 (B) 9 : 4 (C) 2 : 1 (D) 11 : 4

Open solution

Q163. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

The line $2x - y + 1 = 0$ is a tangent to the circle at the point (2, 5) and the centre of the circle lies on $x - 2y = 4$. Then, the radius of the circle is :

(A) $5\sqrt{3}$ (B) $4\sqrt{5}$ (C) $3\sqrt{5}$ (D) $5\sqrt{4}$

Open solution

Q164. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

Choose the incorrect statement about the two circles whose equations are given below :

$x^2 + y^2 - 10x - 10y + 41 = 0$ and

$x^2 + y^2 - 16x - 10y + 80 = 0$

(A) Distance between two centres is the average of radii of both the circles.

(B) Both circles pass through the centre of each other.

(C) Circles have two intersection points.

(D) Both circle's centers lie inside region of one another.

Open solution

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Q165. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

The circle passing through the intersection of the circles, $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 - 4y = 0$, having its centre on the line, $2x - 3y + 12 = 0$, also passes through the point :

(A) $(-3, 1)$ (B) $(1, -3)$ (C) $(-1, 3)$ (D) $(-3, 6)$

Open solution

Q166. JEE Main 2020 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

A circle touches the y-axis at the point (0, 4) and passes through the point (2, 0). Which of the following lines is not a tangent to this circle?

(A) $3x - 4y - 24 = 0$ (B) $4x + 3y - 8 = 0$ (C) $3x + 4y - 6 = 0$ (D) $4x - 3y + 17 = 0$

Open solution

Q167. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1, where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then :

(A) $c^2 + 6c + 7 = 0$ (B) $c^2 - 7c + 6 = 0$ (C) $c^2 - 6c + 7 = 0$ (D) $c^2 + 7c + 6 = 0$

Open solution

Q168. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to :

(A) $\frac{56}{5}$ (B) $\frac{32}{5}$ (C) $\frac{52}{5}$ (D) $\frac{64}{5}$

Open solution

Q169. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

The locus of the centres of the circles, which touch the circle, $x^2 + y^2 = 1$ externally, also touch the y-axis and lie in the first quadrant, is :

(A) $x = \sqrt{1 + 2y}, y \geq 0$ (B) $y = \sqrt{1 + 2x}, x \geq 0$

(C) $y = \sqrt{1 + 4x}, x \geq 0$ (D) $x = \sqrt{1 + 4y}, y \geq 0$

Open solution

Q170. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the circles $x^2 + y^2 + 5Kx + 2y + K = 0$ and $2(x^2 + y^2) + 2Kx + 3y - 1 = 0$, $(K \in R)$, intersect at the points P and Q, then the line $4x + 5y - K = 0$ passes through P and Q, for :

(A) exactly two values of K (B) no value of K

(C) exactly one value of K (D) infinitely many values of K

Open solution

Q171. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The line $x = y$ touches a circle at the point (1,1). If the circle also passes through the point (1, − 3), then its radius is :

(A) 3 (B) 2 (C) $2\sqrt{2}$ (D) $3\sqrt{2}$

Open solution

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Q172. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point :

(A) (6, −2) (B) (4, −2) (C) (−4, 6) (D) (−6, 4)

Open solution

Q173. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If a tangent to the circle $x^2 + y^2 = 1$ intersects the coordinate axes at distinct points P and Q, then the locus of the mid-point of PQ is :

(A) $x^2 + y^2 - 4x^2y^2 = 0$ (B) $x^2 + y^2 - 2xy = 0$

(C) $x^2 + y^2 - 2x^2y^2 = 0$ (D) $x^2 + y^2 - 16x^2y^2 = 0$

Open solution

Q174. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

The tangent and the normal lines at the point $(\sqrt{3}, 1)$ to the circle $x^2 + y^2 = 4$ and the x-axis form a triangle. The area of this triangle (in square units) is :

(A) $\frac{4}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{1}{3}$

Open solution

Q175. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

If a variable line, $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval :

(A) (23, 31) (B) (2, 17) (C) [13, 23] (D) [12, 21]

Open solution

Q176. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

Let C_1 and C_2 be the centres of the circles $x^2 + y^2 - 2x - 2y - 2 = 0$ and $x^2 + y^2 - 6x - 6y + 14 = 0$ respectively. If P and Q are the points of intersection of these circles, then the area (in sq. units) of the quadrilateral PC_1QC_2 is :

(A) 4 (B) 6 (C) 9 (D) 8

Open solution

Q177. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

Two circles with equal radii are intersecting at the points (0, 1) and (0, −1). The tangent at the point (0, 1) to one of the circles passes through the centre of the other circle. Then the distance between the centres of these circles is :

(A) $2\sqrt{2}$ (B) $\sqrt{2}$ (C) 2 (D) 1

Open solution

Q178. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

If a circle C passing through the point (4, 0) touches the circle $x^2 + y^2 + 4x - 6y = 12$ externally at the point (1, − 1), then the radius of C is :

(A) 5 (B) $2\sqrt{5}$ (C) 4 (D) $\sqrt{37}$

Open solution

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Q179. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

If the circles

$x^2 + y^2 - 16x - 20y + 164 = r^2$

and $(x - 4)^2 + (y - 7)^2 = 36$

intersect at two distinct points, then :

(A) r > 11 (B) 0 < r < 1 (C) r = 11 (D) 1 < r < 11

Open solution

Q180. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Three circles of radii a, b, c (a < b < c) touch each other externally. If they have x-axis as a common tangent, then :

(A) a, b, c are in A.P. (B) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P

(C) $\frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}}$ (D) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$

Open solution

Q181. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

If the tangent at (1, 7) to the curve $x^2 = y - 6$

touches the circle $x^2 + y^2 + 16x + 12y + c = 0$, then the value of c is :

(A) 95 (B) 195 (C) 185 (D) 85

Open solution

Q182. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

The tangent to the circle $C_1 : x^2 + y^2 - 2x - 1 = 0$ at the point (2, 1) cuts off a chord of length 4 from a circle C_2 whose center is $(3, -2)$. The radius of C_2 is :

(A) 2 (B) $\sqrt{2}$ (C) 3 (D) $\sqrt{6}$

Open solution

Q183. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

A circle passes through the points (2, 3) and (4, 5). If its centre lies on the line, $y - 4x + 3 = 0$, then its radius is equal to :

(A) 2 (B) $\sqrt{5}$ (C) $\sqrt{2}$ (D) 1

Open solution

Q184. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The radius of a circle, having minimum area, which touches the curve $y = 4 - x^2$ and the lines, y = |x| is :

(A) $2\left(\sqrt{2} - 1\right)$ (B) $4\left(\sqrt{2} - 1\right)$ (C) $4\left(\sqrt{2} + 1\right)$ (D) $2\left(\sqrt{2} + 1\right)$

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Q185. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Equation of the tangent to the circle, at the point (1, −1), whose centre is the point of intersection of the straight lines x − y = 1 and 2x + y = 3 is :

(A) 4x + y − 3 = 0 (B) x + 4y + 3 = 0 (C) 3x − y − 4 = 0 (D) x − 3y − 4 = 0

Open solution

Q186. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

The centres of those circles which touch the circle, $x^2 + y^2 - 8x - 8y - 4 = 0$, externally and also touch the x-axis, lie on :

(A) a circle
(B) an ellipse which is not a circle
(C) a hyperbola
(D) a parabola

Open solution

Q187. JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Let the tangents drawn to the circle, $x^2 + y^2 = 16$ from the point P(0, h) meet the x-axis at points A and B. If the area of ΔAPB is minimum, then h is equal to :

(A) $4\sqrt{3}$ (B) $3\sqrt{3}$ (C) $3\sqrt{2}$ (D) $4\sqrt{2}$

Open solution

Q188. JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the tangent to the conic, $y - 6 = x^2$ at (2, 10) touches the circle

$x^2 + y^2 + 8x - 2y = k$ (for some fixed k) at a point (α, β) ; then (α, β) is :

(A) $\left(-\frac{6}{17}, \frac{10}{17}\right)$ (B) $\left(-\frac{8}{17}, \frac{2}{17}\right)$
(C) $\left(-\frac{4}{17}, \frac{1}{17}\right)$ (D) $\left(-\frac{7}{17}, \frac{6}{17}\right)$

Open solution

Q189. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The number of common tangents to the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$, is :

(A) 3 (B) 4 (C) 1 (D) 2

Open solution

Q190. JEE Main 2014 (Online) 19th April Morning Slot · MCQ (Single Correct Answer)

The equation of the circle described on the chord 3x + y + 5 = 0 of the circle $x^2 + y^2 = 16$ as diameter is :

(A) $x^2 + y^2 + 3x + y - 11 = 0$ (B) $x^2 + y^2 + 3x + y + 1 = 0$
(C) $x^2 + y^2 + 3x + y - 2 = 0$ (D) $x^2 + y^2 + 3x + y - 22 = 0$

Open solution

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Q191. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

For the two circles $x^2 + y^2 = 16$ and $x^2 + y^2 - 2y = 0$, there is/are :

(A) one pair of common tangents
(B) two pairs of common tangents
(C) three common tangents
(D) no common tangent

Open solution

Q192. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

The set of all real values of λ for which exactly two common tangents can be drawn to the circles $x^2 + y^2 - 4x - 4y + 6 = 0$ and
$x^2 + y^2 - 10x - 10y + \lambda = 0$ is the interval :

(A) (12, 32) (B) (18, 42) (C) (12, 24) (D) (18, 48)

Open solution

Q193. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

Let C be the circle with centre at $(1, 1)$ and radius $= 1$. If T is the circle centred at $(0, y)$, passing through origin and touching the circle C externally, then the radius of T is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Open solution

Q194. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

Statement 1: The only circle having radius $\sqrt{10}$ and a diameter along line $2x + y = 5$ is $x^2 + y^2 - 6x + 2y = 0$

Statement 2: $2x + y = 5$ is a normal to the circle $x^2 + y^2 - 6x + 2y = 0$.

(A) Statement 1 is false; Statement 2 is true;
(B) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(C) Statement 1 is true; Statement 2 is false;
(D) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

Open solution

Q195. JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

If a circle C passing through (4, 0) touches the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ externally at a point (1, -1), then the radius of the circle C is:

(A) 5 (B) $2\sqrt{5}$ (C) 4 (D) $\sqrt{57}$

Open solution

Q196. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

The circle passing through $(1, -2)$ and touching the axis of x at $(3, 0)$ also passes through the point :

(A) $(-5, 2)$ (B) $(2, -5)$ (C) $(5, -2)$ (D) $(-2, 5)$

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Q197. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

The number of common tangents of the circles given by $x^2 + y^2 - 8x - 2y + 1 = 0$ and $x^2 + y^2 + 6x + 8y = 0$ is

(A) one (B) four (C) two (D) three

Open solution

Q198. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

The equation of the circle passing through the point $(1, 2)$ and through the points of intersection of $x^2 + y^2 - 4x - 6y - 21 = 0$ and $3x + 4y + 5 = 0$ is given by

(A) $x^2 + y^2 + 2x + 2y + 11 = 0$ (B) $x^2 + y^2 - 2x + 2y - 7 = 0$

(C) $x^2 + y^2 + 2x - 2y - 3 = 0$ (D) $x^2 + y^2 + 2x + 2y - 11 = 0$

Open solution

Q199. AIEEE 2011 · MCQ (Single Correct Answer)

The two circles $x^2 + y^2 = ax$, and $x^2 + y^2 = c^2$ $(c > 0)$ touch each other if :

(A) $|a| = c$ (B) $a = 2c$ (C) $|a| = 2c$ (D) $2|a| = c$

Open solution

Q200. AIEEE 2009 · MCQ (Single Correct Answer)

If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$ then there is a circle passing through P, Q and $(1, 1)$ for :

(A) all except one value of p (B) all except two values of p

(C) exactly one value of p (D) all values of p

Open solution

Q201. AIEEE 2007 · MCQ (Single Correct Answer)

Consider a family of circles which are passing through the point $(-1, 1)$ and are tangent to x-axis. If (h, k) are the coordinate of the centre of the circles, then the set of values of k is given by the interval :

(A) $-\frac{1}{2} \le k \le \frac{1}{2}$ (B) $k \le \frac{1}{2}$

(C) $0 \le k \le \frac{1}{2}$ (D) $k \ge \frac{1}{2}$

Open solution

Q202. AIEEE 2005 · MCQ (Single Correct Answer)

A circle touches the x-axis and also touches the circle with centre at (0, 3) and radius 2. The locus of the centre of the circle is :

(A) an ellipse (B) a circle (C) a hyperbola (D) a parabola

Open solution

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Q203. AIEEE 2004 · MCQ (Single Correct Answer)

A variable circle passes through the fixed point A (p, q) and touches x-axis. The locus of the other end of the diameter through A is :

(A) $(y-q)^2 = 4px$ (B) $(x-q)^2 = 4py$ (C) $(y-p)^2 = 4qx$ (D) $(x-p)^2 = 4qy$

Open solution

Q204. AIEEE 2003 · MCQ (Single Correct Answer)

If the two circles $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct point, then :

(A) $r > 2$ (B) $2 < r < 8$ (C) $r < 2$ (D) $r = 2$.

Open solution

Q205. AIEEE 2002 · MCQ (Single Correct Answer)

The centres of a set of circles, each of radius 3, lie on the circle $x^2 + y^2 = 25$. The locus of any point in the set is :

(A) $4 \le x^2 + y^2 \le 64$ (B) $x^2 + y^2 \le 25$

(C) $x^2 + y^2 \ge 25$ (D) $3 \le x^2 + y^2 \le 9$

Open solution

Q206. AIEEE 2002 · MCQ (Single Correct Answer)

The centre of the circle passing through (0, 0) and (1, 0) and touching the circle $x^2 + y^2 = 9$ is :

(A) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, -\sqrt{2}\right)$

(C) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{3}{2}\right)$

Open solution

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Answer Key

Numerical

1. **18**	2. **18**	3. **80**	4. **8**	5. **2**
6. **46**	7. **768**	8. **19**	9. **9**	10. **1575**
11. **2**	12. **100**	13. **7**	14. **2**	15. **3**
16. **121**	17. **24**	18. **72**	19. **27**	20. **12**
21. **10**	22. **16**	23. **7**	24. **7**	25. **165**
26. **16**	27. **56**	28. **3**	29. **7**	30. **3**
31. **36**				

MCQ (Single Correct Answer)

32. **C**	33. **B**	34. **C**	35. **B**	36. **C**
37. **D**	38. **D**	39. **D**	40. **D**	41. **C**
42. **A**	43. **C**	44. **C**	45. **C**	46. **B**
47. **C**	48. **D**	49. **B**	50. **A**	51. **A**
52. **B**	53. **A**	54. **C**	55. **D**	56. **B**
57. **A**	58. **A**	59. **C**	60. **D**	61. **D**
62. **C**	63. **B**	64. **C**	65. **A**	66. **D**
67. **D**	68. **A**	69. **A**	70. **A**	71. **B**
72. **B**	73. **D**	74. **A**	75. **A**	76. **D**
77. **A**	78. **D**	79. **B**	80. **D**	81. **C**
82. **D**	83. **B**	84. **B**	85. **A**	86. **B**
87. **B**	88. **C**	89. **D**	90. **C**	91. **B**
92. **A**	93. **D**	94. **B**	95. **B**	96. **C**
97. **B**	98. **A**	99. **B**	100. **C**	101. **B**
102. **B**	103. **B**	104. **D**	105. **B**	106. **C**
107. **B**	108. **A**	109. **D**	110. *Scan QR*	111. **B**
112. **A**	113. **D**	114. **B**	115. **A**	116. **A**
117. **A**	118. **A**	119. **C**	120. **D**	121. **D**
122. **B**	123. **D**	124. **D**	125. **B**	126. **D**
127. **D**	128. **D**	129. **C**	130. **C**	

Out of syllabus

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131. **11**	132. **3**	133. **25**	134. **816**	135. **11**
136. **18**	137. **40**	138. **61**	139. **1**	140. **1**
141. **1225**	142. **9**	143. **C**	144. **A**	145. **C**
146. **B**	147. **B**	148. **C**	149. **B**	150. **A**
151. **D**	152. **C**	153. **C**	154. **B**	155. **D**
156. **C**	157. **C**	158. **C**	159. **C**	160. **C**
161. **B**	162. **B**	163. **C**	164. **D**	165. **D**
166. **B**	167. **A**	168. **D**	169. **B**	170. **B**
171. **C**	172. **A**	173. **A**	174. **C**	175. **D**
176. **A**	177. **C**	178. **A**	179. **D**	180. **C**
181. **A**	182. **D**	183. **A**	184. **B**	185. **B**
186. **D**	187. **D**	188. **B**	189. **A**	190. **A**
191. **D**	192. **B**	193. **B**	194. **A**	195. **A**
196. **C**	197. **C**	198. **D**	199. **A**	200. **A**
201. **D**	202. **D**	203. **D**	204. **B**	205. **A**
206. **B**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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